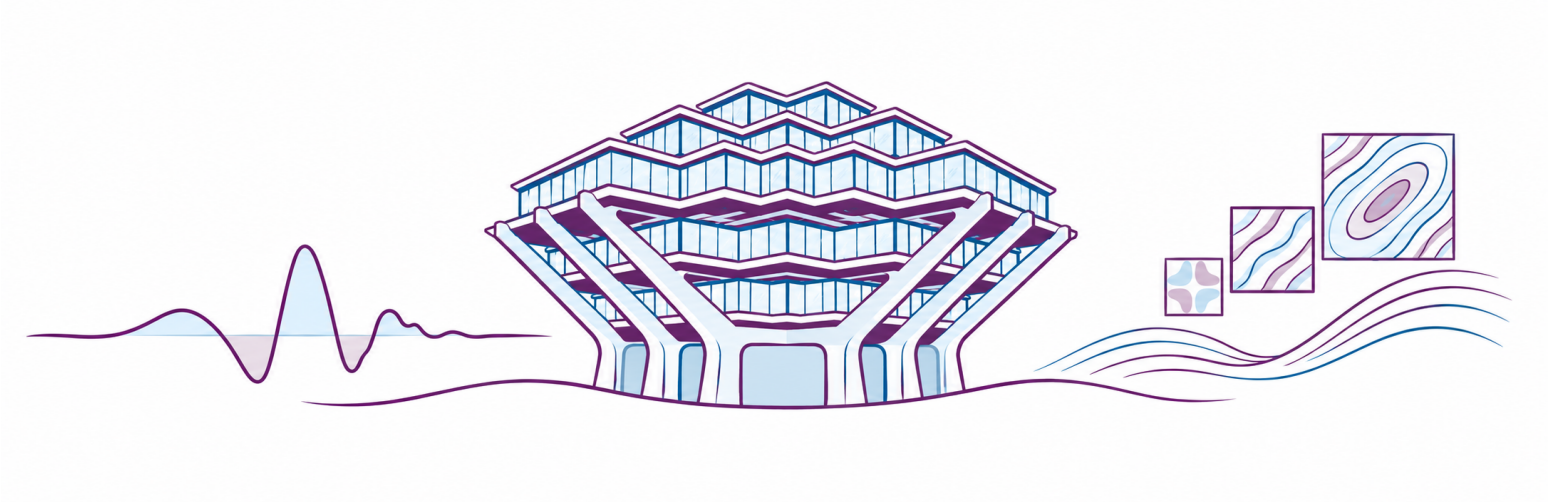
ECE 251C Lecture 14

Discrete Wavelet Transforms and Hard Thresholding

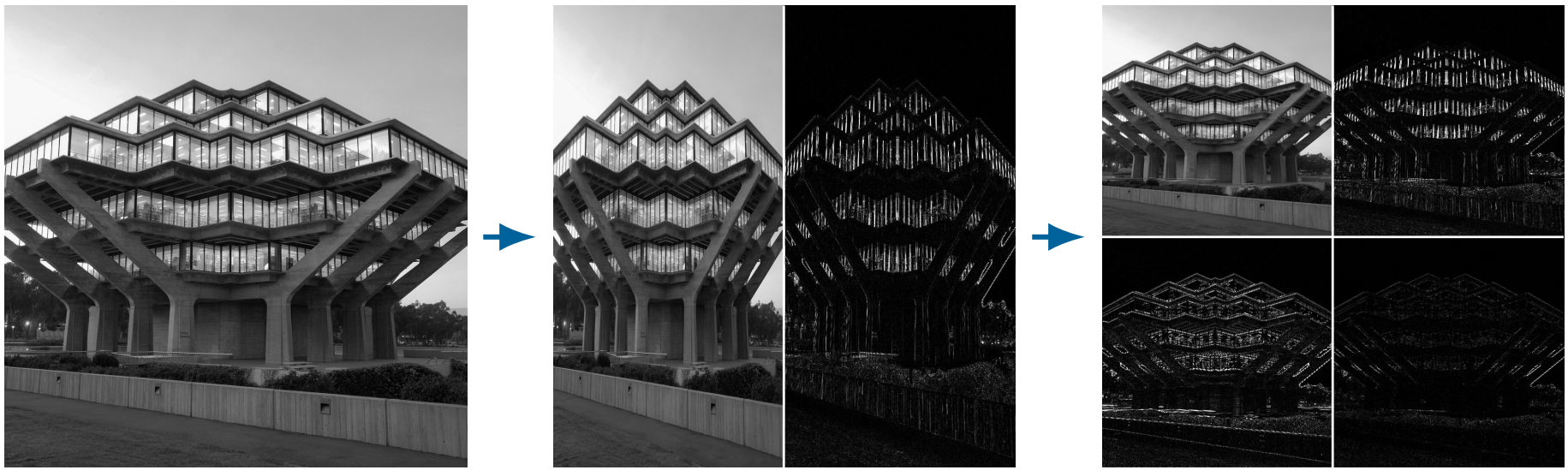
Rahul Parhi

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University of California, San Diego



Filtering Rows, Then Columns

Image | **Rows** | **Columns**



1024×1024 | Two 1024×512 arrays | Four 512×512 arrays

Low-pass and high-pass filtering, followed by downsampling by 2.

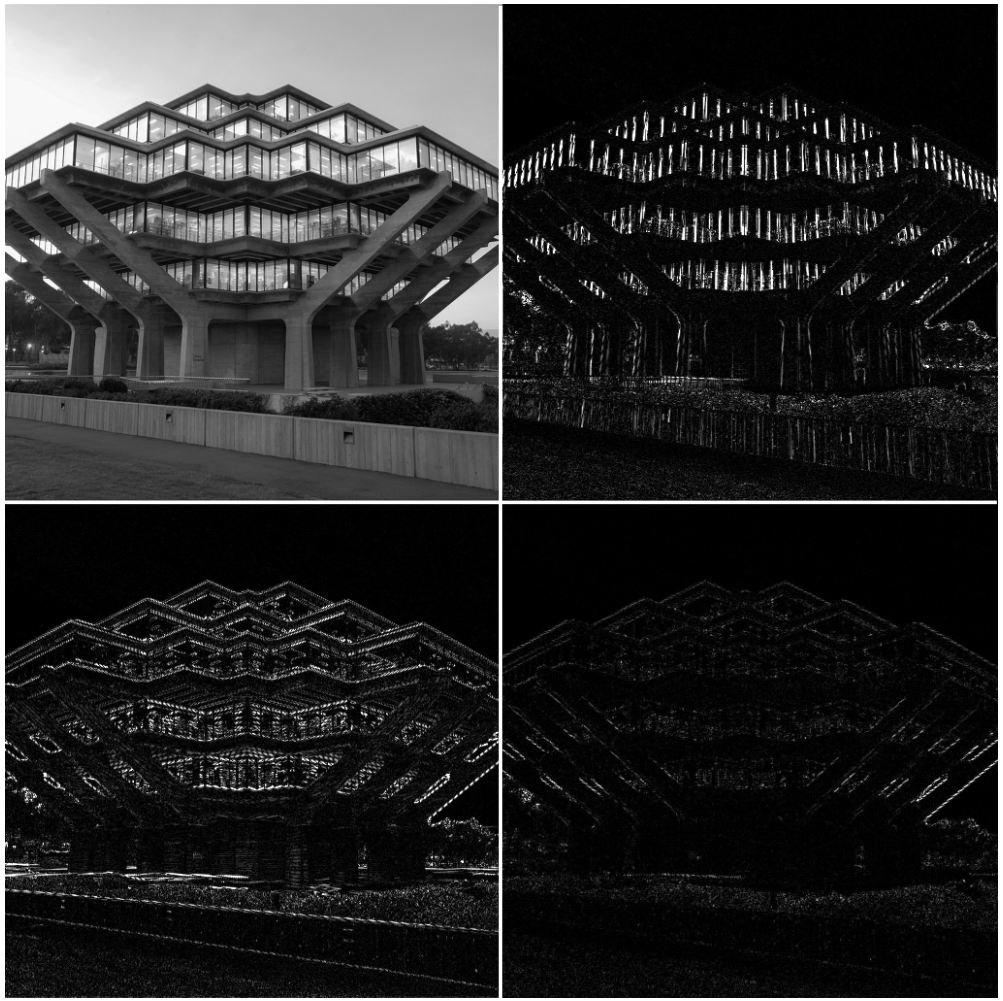
$$4(512 \times 512) = 1024 \times 1024 \text{ coefficients}$$

The Four Subbands

The first letter describes row filtering; the second describes column filtering.

LL

Approximation



HL

Vertical details

LH

Horizontal details

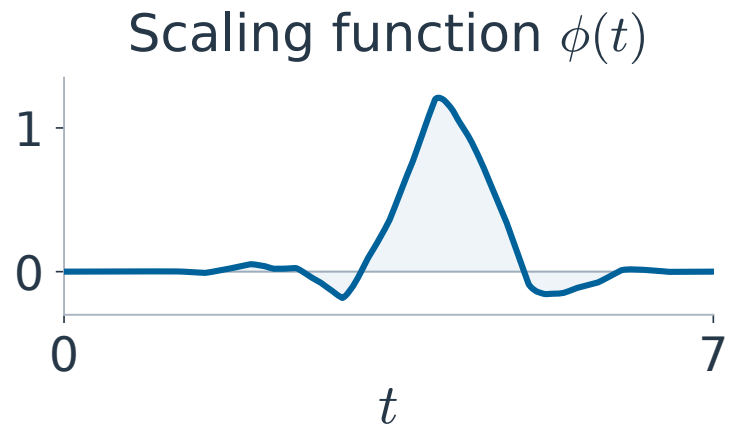
HH

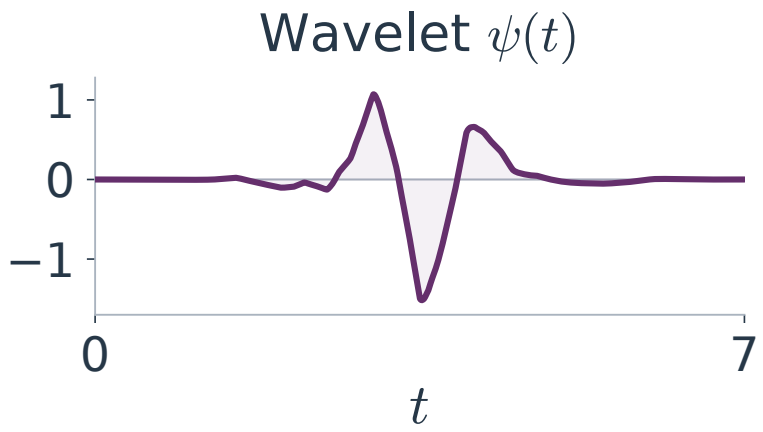
Diagonal details

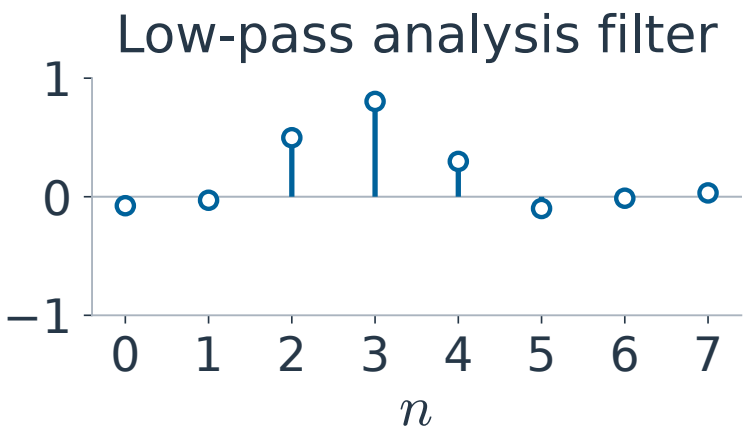
Detail magnitudes are amplified for display.

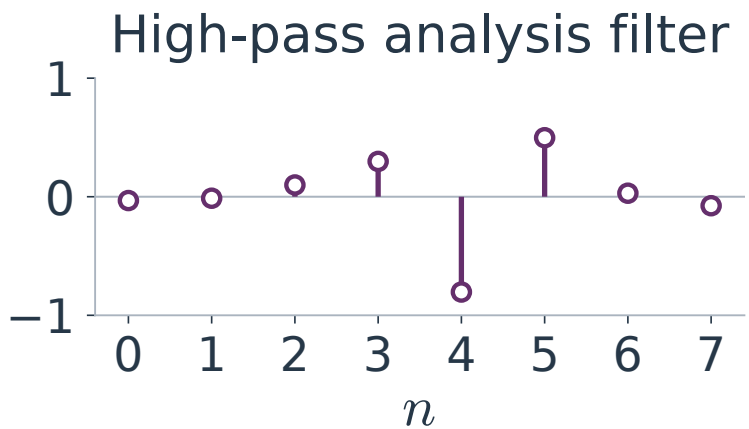
The Wavelets Used Here

Symmlet-4 has **8 filter taps** and **4 vanishing moments**.



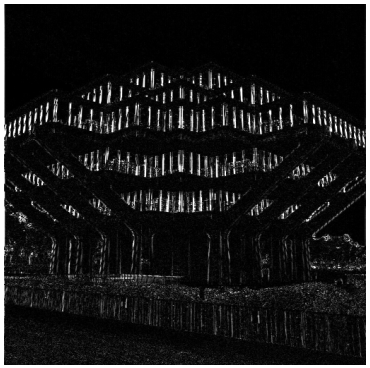


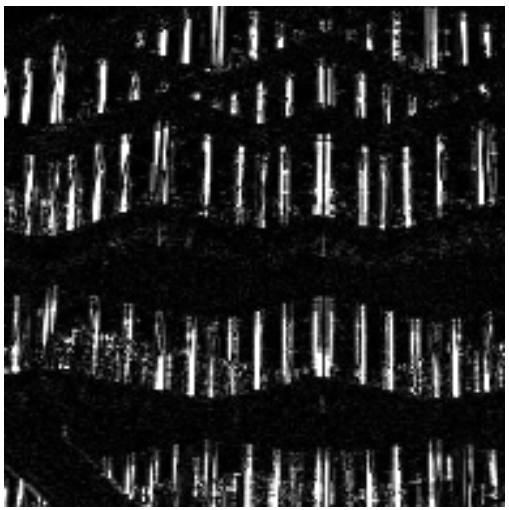




Directional Details

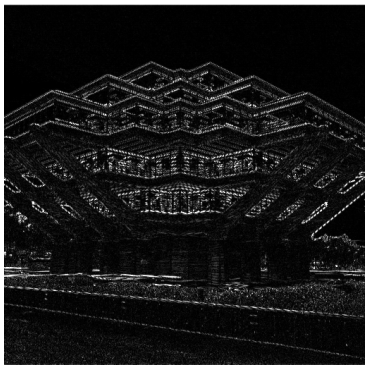
HL

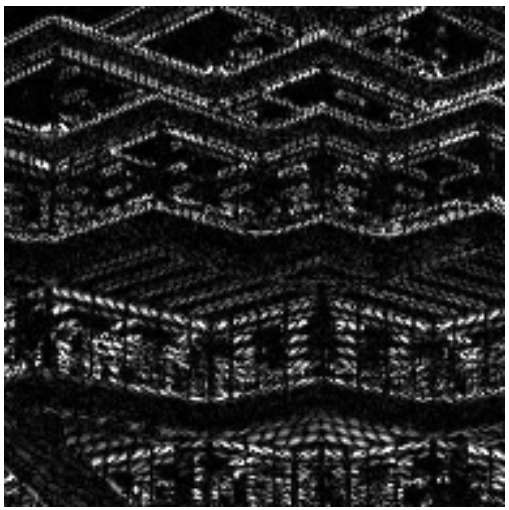




High-pass along rows
Vertical boundaries

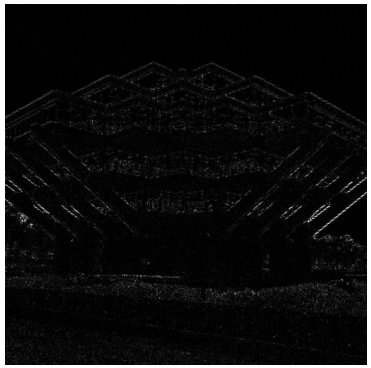
LH

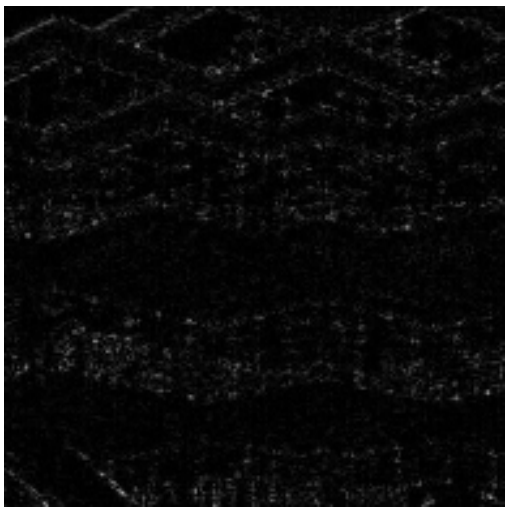




High-pass along columns
Horizontal boundaries

HH



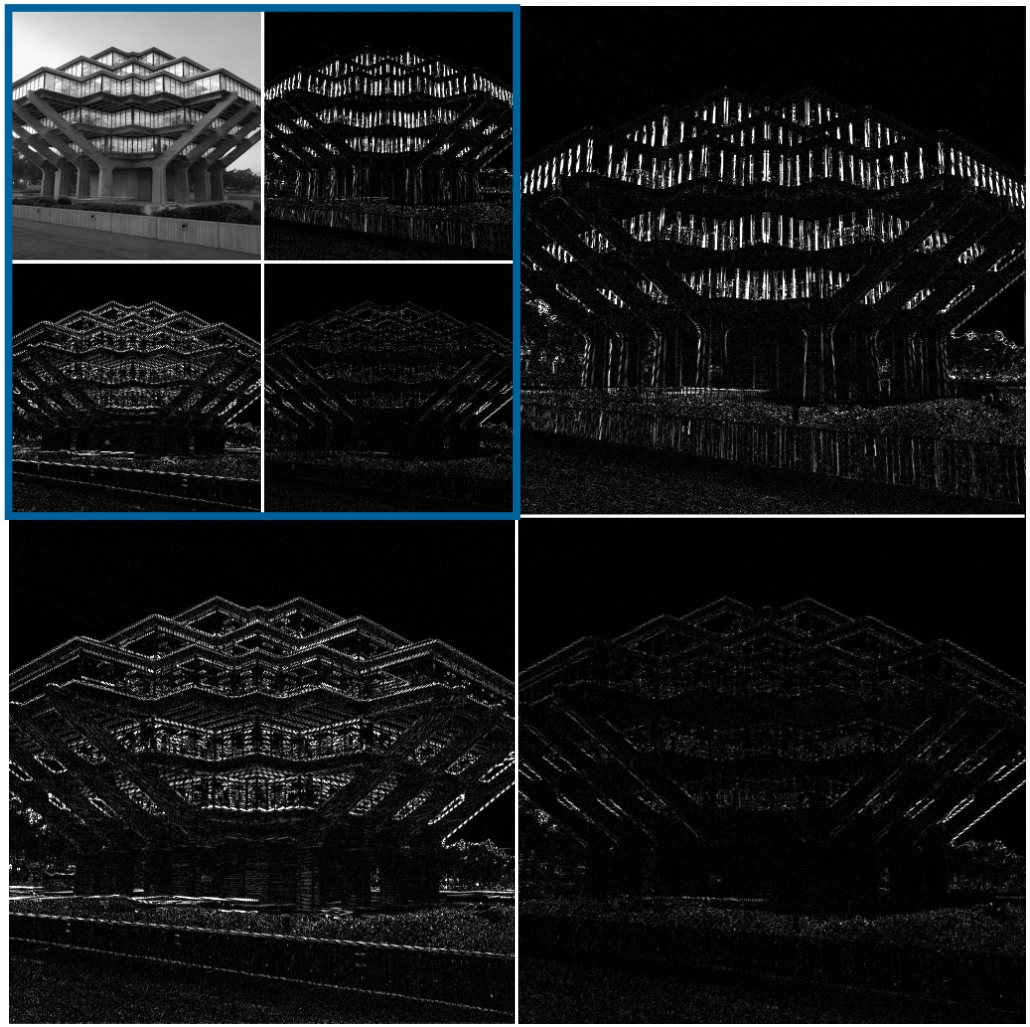


High-pass in both directions
Diagonal details

The lower images enlarge the same region of each subband.

The Next Stage Acts on LL

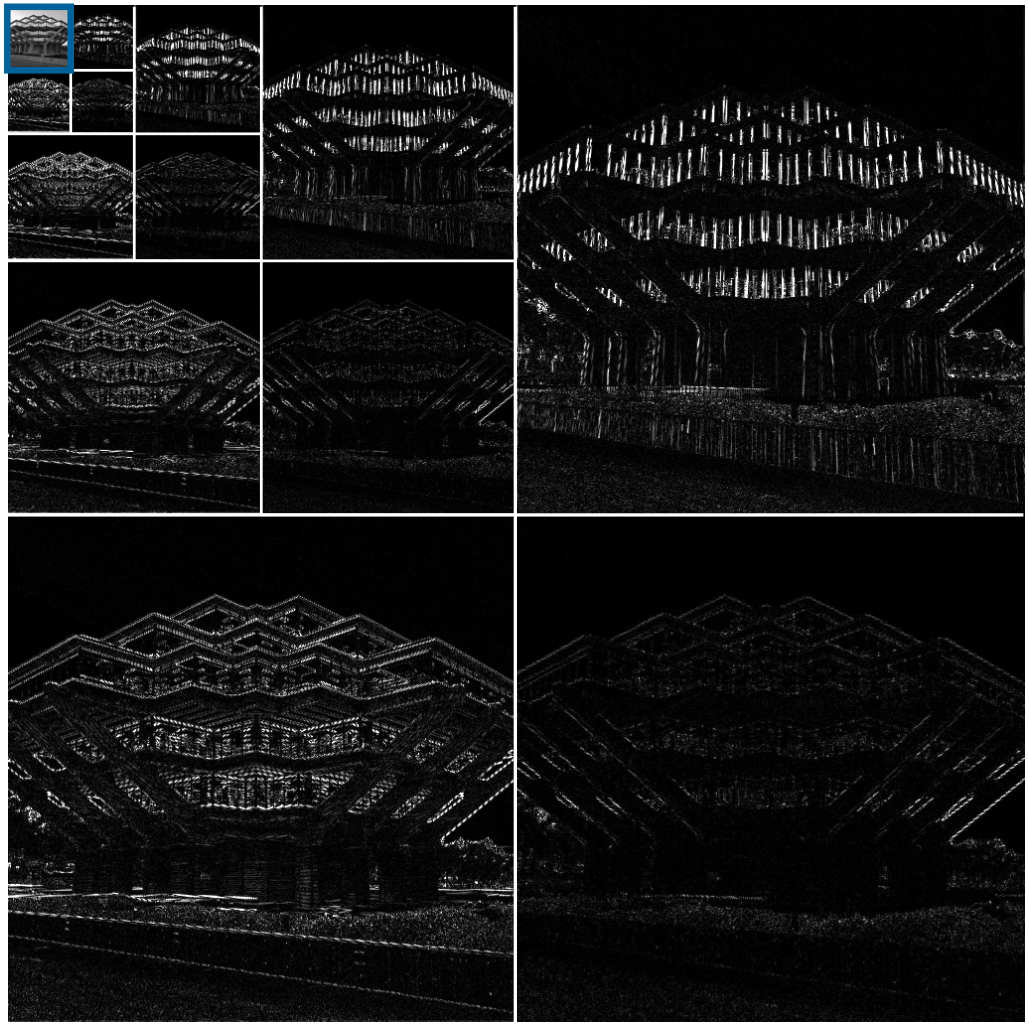
Apply the same four-way split



The detail subbands stay in place. Only LL is decomposed further.

Four Levels of the 2D DWT

Coarsest approximation: 64×64



1,048,576 pixels $\longleftrightarrow$ **1,048,576 coefficients**

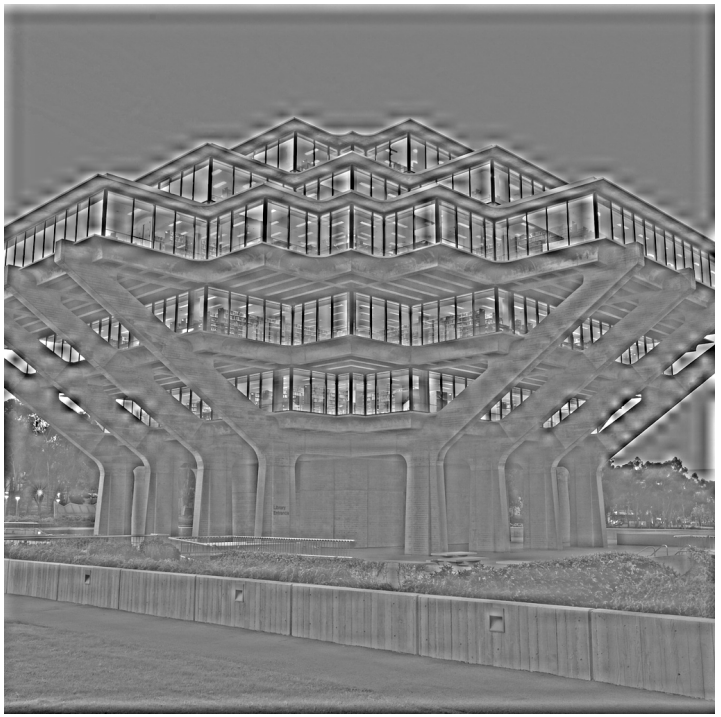
Coarse Approximation and Detail

Coarsest LL only



4,096 approximation coefficients

All detail subbands



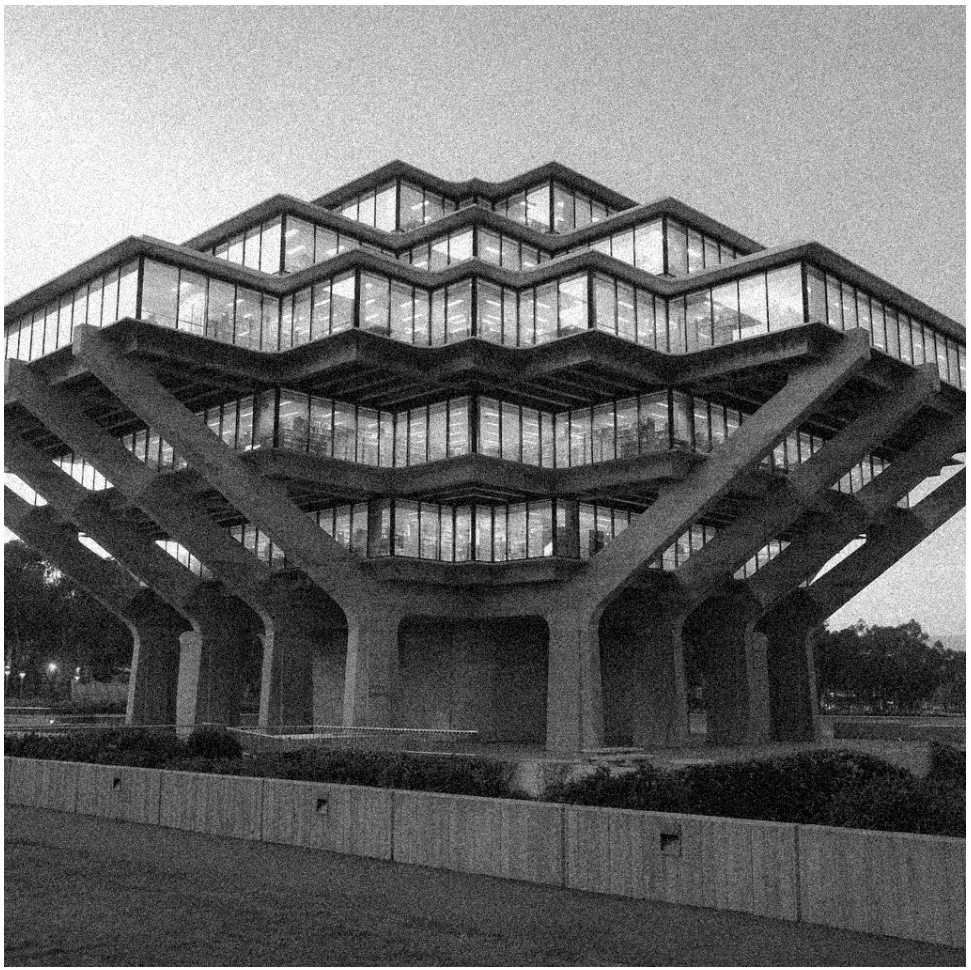
Signed contribution; contrast enhanced

$$x[n, m] = x_{\text{coarse}}[n, m] + x_{\text{detail}}[n, m].$$

Keeping every coefficient gives perfect reconstruction.

A Noisy Observation

$$y = x + \varepsilon, \quad \varepsilon[n, m] \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma = 0.08, \quad x[n, m] \in [0, 1].$$



For an estimate $\bar{x}$ of x:

Mean squared error

$$\text{MSE} = \frac{\|\bar{x} - x\|_{\ell^2}^2}{N}$$

$N = 1024^2$ pixels

Peak signal-to-noise ratio

$$\text{PSNR} = 10 \log_{10}\left(\frac{1}{\text{MSE}}\right) \text{ dB}$$

Higher PSNR means smaller error.

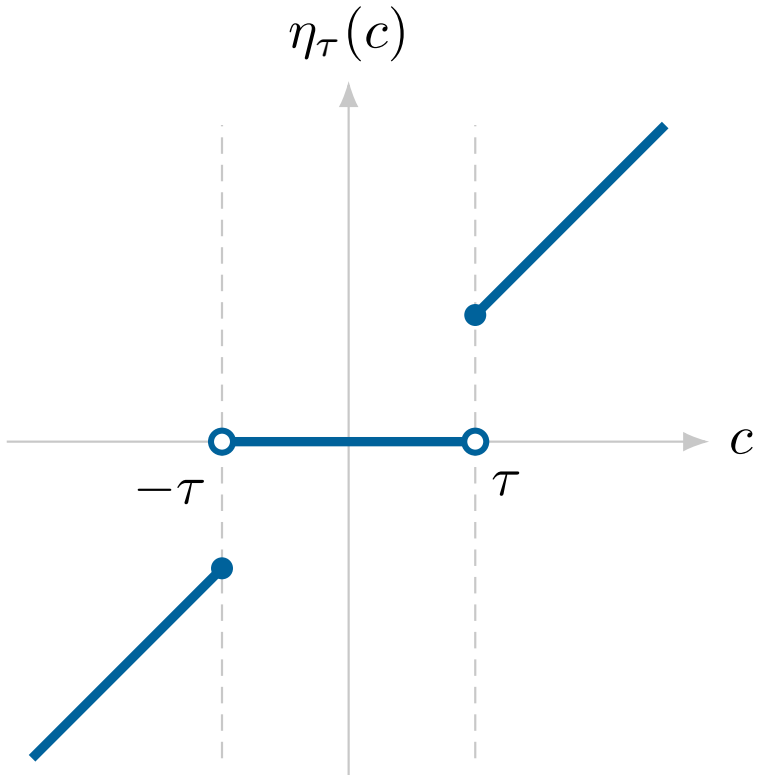
Noisy image $(\bar{x} = y)$ PSNR: **21.93 dB**

Hard Thresholding

$$\eta_\tau(c) = \begin{cases} c, & |c| \geq \tau, \\ 0, & |c| < \tau. \end{cases}$$

Retain large coefficients.

Set small coefficients to zero.



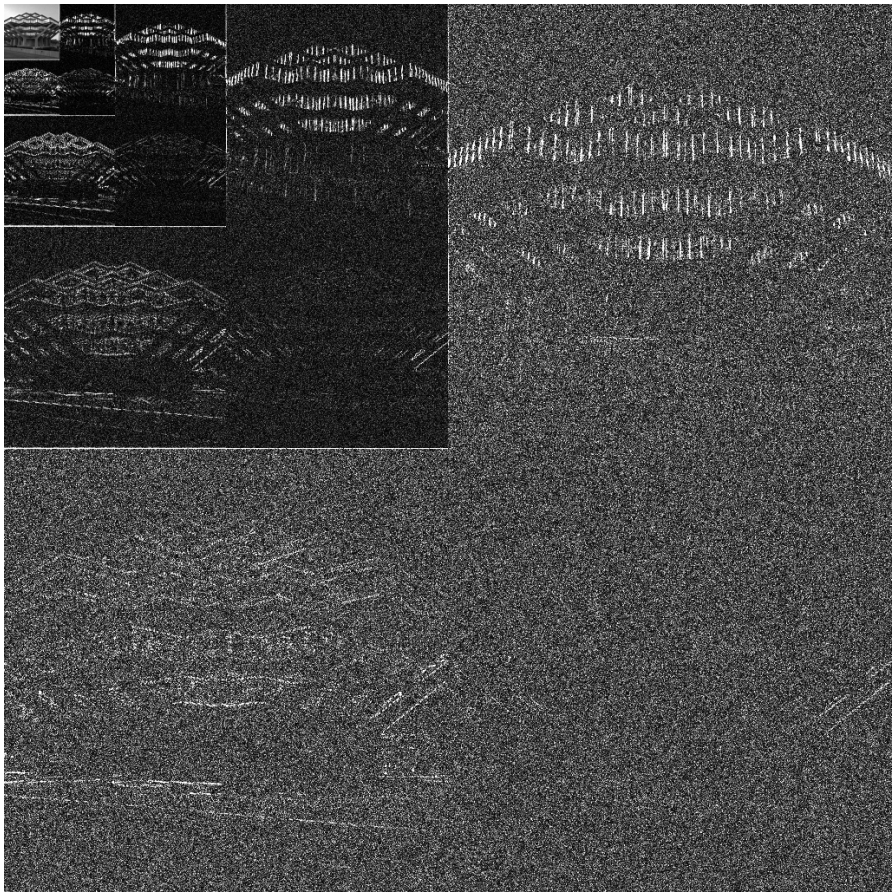
$$\boxed{\bar{x} = \mathrm{T}^{\mathsf{T}} \eta_\tau(\mathrm{T}y)}, \qquad \mathrm{T}^{\mathsf{T}}\mathrm{T} = \mathrm{I}.$$

T: orthonormal wavelet or real Fourier transform.

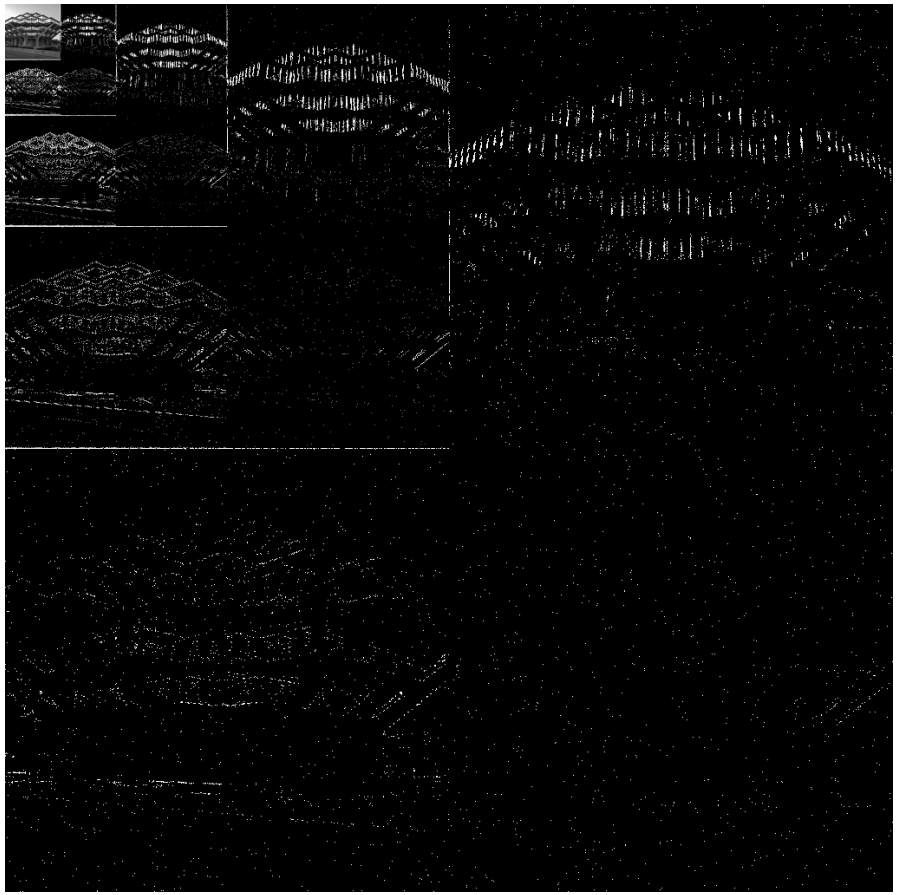
The same threshold in both bases: $\tau = 3\sigma = 0.24$.

Hard Thresholding in the Wavelet Domain

Noisy coefficients



After hard thresholding



The same display gain is used before and after thresholding.

29,119 of 1,048,576 coefficients survive 2.78%

Wavelet Denoising

Four-level symmlet-4 DWT, followed by hard thresholding at 3σ.

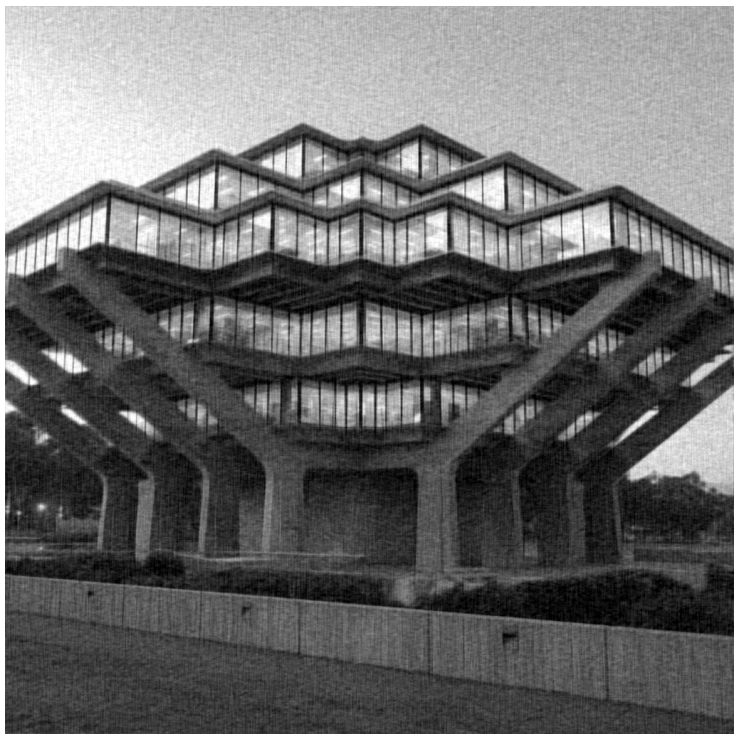


Wavelet reconstruction PSNR: **28.04 dB**

Fourier and Wavelet Denoising

Identical noisy input and threshold $\tau = 3\sigma$ in both real orthonormal bases.

Fourier



26.29 dB PSNR

3.09% of coefficients retained

Wavelet



28.04 dB PSNR

2.78% of coefficients retained

Wavelets give **33% lower mean squared error** for this image.

A Closer Look at the Same Region

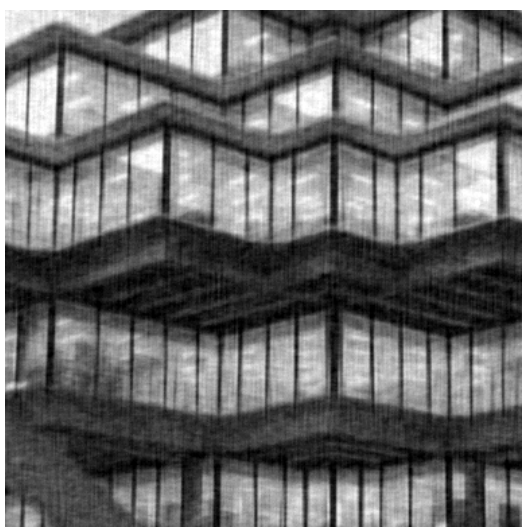
Reference



Noisy



Fourier

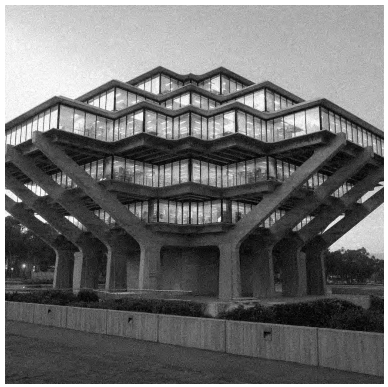


Wavelet



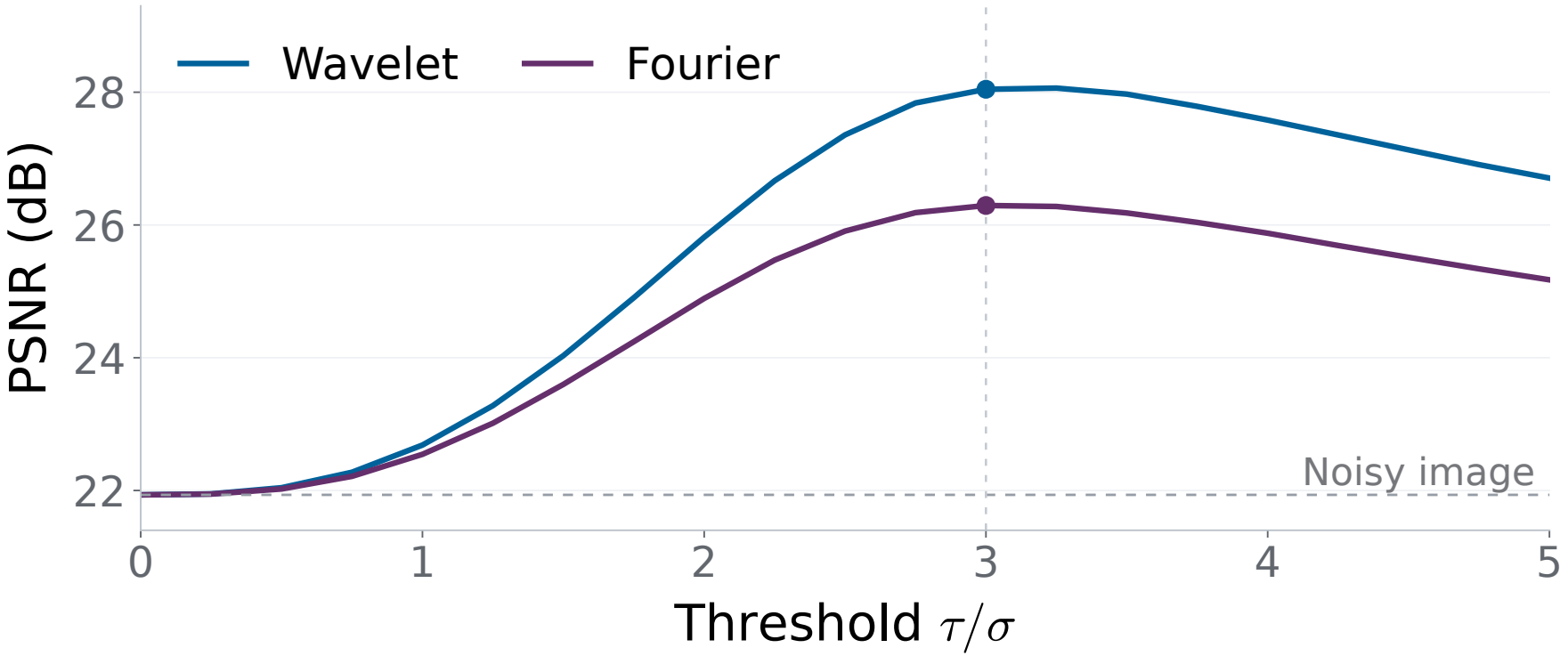
The Effect of the Threshold

Smaller thresholds retain more noise. Larger thresholds remove more image detail.

$\tau = 1.5\sigma$	$\tau = 3\sigma$	$\tau = 5\sigma$
		
		
24.03 dB PSNR	**28.04 dB PSNR**	**26.71 dB PSNR**

Denoising Quality and the Threshold

Higher PSNR means lower mean squared error.



The reference image is used only to measure the error.